

# ITIL

## Systems Development and Acquisition

Systems Development and Acquisition is defined by the Australian Computer Society (ACS) as "An understanding is required of how to develop or acquire software (information) systems that satisfy the requirements of users and customers. All phases of the lifecycle of an information system should be understood including: requirement analysis (systems analysis) and specification, design, construction, testing, and operation and maintenance. There should also be knowledge of methodologies and processes for developing systems."

ACS also defines Systems Software (SYSP) as:

"The provision of specialist expertise to facilitate and execute the installation and maintenance of system software such as operating systems, data management products, office automation products and other utility software."

## IT service management

(ITSM) refers to the entirety of activities – directed by policies, organized and structured in processes and supporting procedures – that are performed by an organization to design, plan, deliver, operate and control information technology (IT) **services** offered to customers.

**ITSM - What is it? Introduction to IT Service Management (<https://youtu.be/Zt6GNAIoUsY>)**

Most organisations will have some sort of IT department, responsible for the management of hardware and software, network infrastructure, and in-house software tools and software development. Once software has been specified, developed and deployed, it must be maintained. But this is only a part of what contemporary IT departments do.

IT departments also acquire, purchase, lease and outsource software and software development, maintain relationships with IT vendors, IT users and management. IT departments are at the “front line” of dealing with cyber-attacks, breaches, privacy violations, compliance with regulations, and strategic decisions which affect the technology of the entire business.

IT Service Management (ITSM) is the management of all activities related to IT through the use of policies, procedures and standards. IT Service Management is different to other types of management (network management, identity and access management, systems management) because its focus is on processes which improve the services provided to “customers” – usually the computer users and IT providers of a business. The main foci are supporting the “customer experience” and improving service quality.

## ITSM

ITSM borrows processes from the allied fields of information security management and software engineering.

A primary function of ITSM is maintaining and operating an IT service desk (or help desk or call centre) which is a single point of contact (SPOC) for communicating with IT users and stakeholders or “customers”.

Activities performed by ITSM include referrals to an engineering department (who deliver and maintain IT systems), network administration (who maintain network infrastructure), system admin. (who maintain software and computers), desktop support (who fix computers), account and identity management (who maintain access controls and user accounts), System support (who maintain data centres and back-end services) and acquisition (who purchase/lease services from 3<sup>rd</sup> parties and manage tendering processes for purchasing new systems). ITSM will also have a cadre of software developers who develop middleware which enables the many diverse systems being used to share information and interact with each other.

ITSM SPOCs generally use ticketing systems and SysOps monitoring to gather and organise information on the functioning of IT services. The service desk will respond to “Tickets” raised by customers to prioritise and schedule system maintenance, acquisition of software or software licences, create accounts, and other maintenance tasks. A SysOps team will monitor system and network use and troubleshoot system problems (such as spikes in latency, bottlenecks, cyberattacks) and refer issues to specific teams for remediation. When problems become insurmountable recommendations to management for new systems/architectures or services are made to management.

Watch this video on what ITSM actually is:

**[What is ITIL and IT Service Management \(ITSM\)? \(https://youtu.be/ivKtKI7GKjg\)](https://youtu.be/ivKtKI7GKjg)**

There are various frameworks for ITSM, or aspects of ITSM (COBIT, ISO2000, ISO27002, ISO9000, COSO) but the dominant one is ITIL: Information Technology Infrastructure Library.

# ITIL

ITIL is a set of ITSM practices which “align” IT services with business needs. ITIL is a commercial product, which allows business to licence the ITIL intellectual property (checklists, tasks, processes, procedures).

Various versions of ITIL exist, the current version being version 4.

The ITIL 4 Foundation provides certification for individuals who complete ITIL 4 training and assessments. Certification standards are managed by the ITIL Certification Management Board (ICMB). Personal cITIL certification is owned by Axelos (<https://en.wikipedia.org/wiki/AXELOS>), with various licensed bodies able to run accredited training and assessment.

There are various levels of ITIL certification, each of which provide an income stream to Axelos and training providers.

Source: <https://cxpgo.quintica.com/itil-3/> (<https://cxpgo.quintica.com/itil-3/>)

## Advantages of ITIL certification

- Well-recognised certification scheme for individuals working in IT Service Management
- Alignment between IT and business
- Improvements in service delivery and customer satisfaction
- reduced costs through visibility of IT costs and assets
- Management of business risk and service disruption
- A service environment which supports constant change
- Possible integration with some DevOps tools

## Disadvantages of ITIL certification

- Only available for individuals, not organisations
- Expensive and proprietary ITIL resources (not available outside of paid training)
- Specific training required for certification and implementation
- ITIL not directly aligned with ITSM frameworks such as ISO2000

- Outdated definitions and terminology
- Overdependence on strict compliance and doing things “by the (5) books”
- ITIL requires practitioners to be involved in many overlapping areas, causing duplication of work rather than specialisation.
- The effectiveness of ITIL is disputed (<https://web.archive.org/web/20050404165524/http://www.cio.com/leadership/buzz/column.html?ID=4186> (<https://web.archive.org/web/20050404165524/http://www.cio.com/leadership/buzz/column.html?ID=4186>)).

ITIL has been described by various organisations (both in favour of and against ITIL) as a “philosophy”, an “cult” or a “religion”. (<http://www.itskeptic.org/node/> (<http://www.itskeptic.org/node/17>)).

## ITIL has 5 stages

(or “books”) and (depending on the version) up to 26 process areas.

1. Service strategy
2. Service Design
3. Service Transition
4. Service Operation
5. Continual Service Improvement

## ITIL has three main components:

1. SPOC (service/help desk)
2. Service Management (descriptions of service capabilities)
3. Service Catalogues (details of services available to customers and internal staff)

Some ITIL terminology:

- Request for Change (RFC) – formal process for actions by IT
- Change Advisory Board (CAB) – decide which requested changes will be implemented
- Incident – an unusual event
- Problem – a series of incidents of the same type
- SLA – Service level agreement – an agreement between IT and the customer
- OLA – An SLA between IT and an external organisation

Service Strategy: Designed for annual, bi-annual reviews or planning sessions

1. Portfolio management – which services will we keep, which will we add or remove?
2. Financial management and Return on Investment – profitable or not?
3. Demand Management – who wants it and can we deliver it?

### Service Design: Which services will we offer?

4. Catalog management
5. Service level management
6. Supplier management
7. Capacity management
8. Availability management
9. IT Service continuity management
10. InfoSec management

Issues include which technologies (e.g. browsers) will be supported, who has the knowledge and capability to support things and what happens during a cybersecurity incident.

### Service Transition: Manage training, change and expectations

11. Change management
12. Service asset and configuration management
13. Release and deployment management
14. Service validation and testing evaluations
15. Knowledge management

### Service Operation: Management of existing resources

16. Event management
17. Incident management
18. Request management
19. Problem management
20. Access management

Note that there is a lot of duplication of expertise across service design, transition and operation. The use of a ticketing system and effective communications between areas is essential to manage the handover of system information from one group to the next.

The parts of the organisation involved include the service desk, technical management, application management and operations.

### Continuous service improvement: A 7-step plan for improvement

1. Define desirable metrics
2. Define actual metrics
3. Measure
4. Process data
5. Analyse data
6. Present information
7. Implement corrective action

ITIL is really big on poster-size wall charts with many words in different-shaped boxes. [.\(https://www.brighttalk.com/webcast/534/354500/is-til-4-devops-ready\)](https://www.brighttalk.com/webcast/534/354500/is-til-4-devops-ready)



## Live online seminar

**Seminar Slides** (<https://swinburnesarawak.instructure.com/courses/2164/files/1640076?wrap=1>) [↓](https://swinburnesarawak.instructure.com/courses/2164/files/1640076/download?download_frd=1)  
([https://swinburnesarawak.instructure.com/courses/2164/files/1640076/download?download\\_frd=1](https://swinburnesarawak.instructure.com/courses/2164/files/1640076/download?download_frd=1))

## COS40006 - Week 8 Webinar - ITIL & CapEx

